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| 7 (a) minimum energy to remove an electron from the metal/surface | B1 | [1] |
| (b) gradient = 4.17×10^{-15} (allow 4.1 $\rightarrow$ 4.3) | | |
| $h = 4.15 \times 10^{-15} \times 1.6 \times 10^{-19}$ or $h = 4.1 \text{ to } 4.3 \times 10^{-15} \text{ eVs}$ | C1 | |
| $= 6.6 \times 10^{-34} \text{ Js}$ | A1 | |
| | A0 | [2] |
| (c) graph: straight line parallel to given line | | |
| with intercept at any higher frequency | B1 | |
| intercept at between $6.9 \times 10^{14} \text{ Hz}$ and $7.1 \times 10^{14} \text{ Hz}$ | B1 | [3] |

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8 (a) nuclei having same number of protons/proton (atomic) number

P1